Semester II **B. Tech CSE (AI)**

BCSAI 203: DESIGN THINKING

3L + 0T + 0P + 3C **MM 100**

Unit 1: Design Thinking: Foundations of Human Centered Design, Barriers to Innovation and Adoption, Learning by Doing, Understanding Needs in Context.

Unit 2: Design Needs and Interventions: The Ethics of Design Interventions, Design Needs in Education, Engineering and Health & Society.

Unit 3: Empathy in Design: Discovering Explicit and Latent Needs, Qualitative Research: Watching and Listening, Point of View & Problem Reframing, Developing Grounded Theory, Design for Usability.

Unit 4: Ideation, Experimentation and Evolution: Generating and Developing Ideas, Creativity as Teaching and Learning, Prototyping and testing Learning Through Things & Interactions, Express, Test, Cycle

Unit 5: Design Documentation: Representing Design Knowledge, Diffusion of Innovation, Design as research

Text/Reference books:

1. Design Thinking: Integrating Innovation, Customer Experience, and Brand Value,byThomas Lockwood,Allworth Press, 2010.
2. 101 Design Methods: A Structured Approach for Driving Innovation in Your Organization Book by Vijay Kumar.
3. Design Thinking: Understand – Improve – Apply edited by Hasso Plattner, Christoph Meinel, Larry Leifer, Springer Science & Business Media, 201